EE-421: LAB - Lab Work #2

Analysis and Design of Basic Finite State Machines (FSMs)

UMBB / IGEE / DE / M01 / C.E

Part 1: FSM Analysis

1. Analysis of Figure 1 (Mealy Model FSM)

A. Circuit Equations from Schematic

From the schematic of **Fig. 1**:

- Top OR gate for D_1:
$$D_1 = w \cdot Q_0 + w \cdot Q_1$$
- Bottom OR gate for D_0:
$$D_0 = w \cdot \overline{Q_0} + w \cdot Q_1$$
- Output gate for y:
$$y = \overline{w} \cdot Q_1 \cdot Q_0$$

State assignment: $A = 00$, $B = 01$, $C = 10$, $D = 11$.

B. Excitation Table

Format: **Input** — **Present State** — **Next State** — **FF Inputs** — **Output** (Ordered with all $w = 0$ rows first, then all $w = 1$ rows).

Table 1: Excitation Table for Fig. 1 (Mealy)

Input (w)	**Present State**		**Next State**		**FF Inputs**		**Output** (y)
	Q_1	Q_0	Q_1^+	Q_0^+	D_1	D_0	
0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0
0	1	0	0	0	0	0	0
0	1	1	0	0	0	0	1
1	0	0	0	1	0	1	0
1	0	1	1	0	1	0	0
1	1	0	1	1	1	1	0
1	1	1	1	1	1	1	0

Table 2: State Table for Fig. 1 (Mealy)

w	**P.S**	**N.S**	y
0	A	A	0
0	B	A	0
0	C	A	0
0	D	A	1
1	A	B	0
1	B	C	0
1	C	D	0
1	D	D	0

C. Equation Validation using K-Maps

K-Map for $D_1(w, Q_1, Q_0)$ (Horizontal Layout):

w \ Q_1Q_0	00	01	11	10
0	0	0	0	0
1	0	1	1	1

- Group 1 (wQ_0): Combines cells at $wQ_1Q_0 = 101, 111$.
- Group 2 (wQ_1): Combines cells at $wQ_1Q_0 = 110, 111$.
- **Validated Equation:** $D_1 = wQ_0 + wQ_1$ ✓

K-Map for $D_0(w, Q_1, Q_0)$ (Horizontal Layout):

w \ Q_1Q_0	00	01	11	10
0	0	0	0	0
1	1	0	1	1

- Group 1 ($w\overline{Q_0}$): Combines cells at $wQ_1Q_0 = 100, 110$.
- Group 2 (wQ_1): Combines cells at $wQ_1Q_0 = 110, 111$.
- **Validated Equation:** $D_0 = w\overline{Q_0} + wQ_1$ ✓

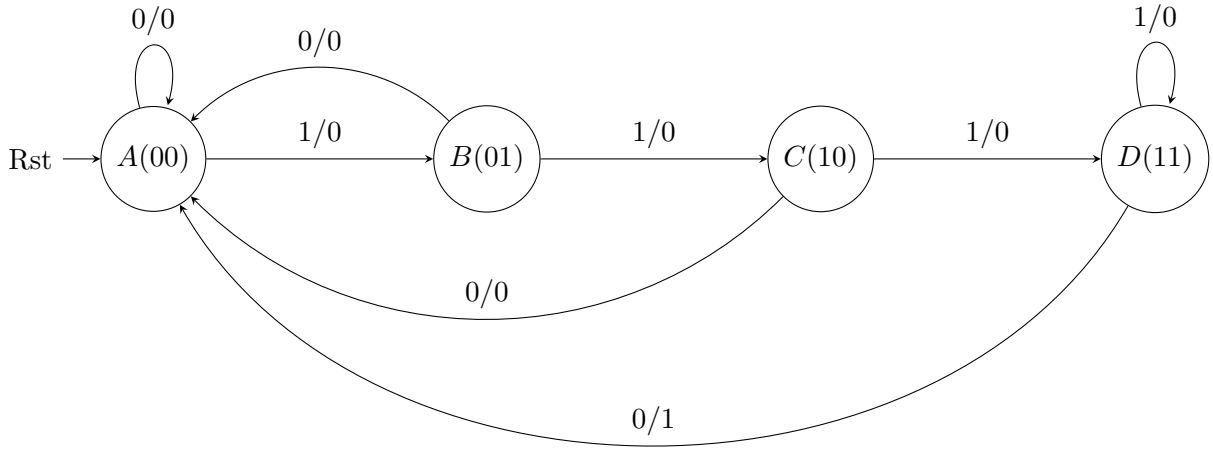
K-Map for Output $y(w, Q_1, Q_0)$ (Horizontal Layout):

w \ Q_1Q_0	00	01	11	10
0	0	0	1	0
1	0	0	0	0

- Single cell at $w = 0, Q_1Q_0 = 11$: $\overline{w} \cdot Q_1 \cdot Q_0$
- **Validated Equation:** $y = \overline{w}Q_1Q_0$ ✓

The FSM is a Mealy model which detects the sequence "1110".

D. State Diagram (Mealy Model Layout)



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2. Analysis of Figure 2 (Moore Model FSM)

A. Circuit Equations from Schematic

From the schematic of **Fig. 2**:

- Top OR gate for D_1:
$$D_1 = w \cdot \overline{Q_0} \cdot Q_1 + \overline{w} \cdot Q_0$$
- Bottom connection for D_0:
$$D_0 = w$$
- Output gate for y:
$$y = Q_1 \cdot Q_0$$

B. Excitation Table

Format: **Input** — **Present State** — **Next State** — **FF Inputs** — **Output** (Ordered with all $w = 0$ rows first, then all $w = 1$ rows).

Table 3: Excitation Table for Fig. 2 (Moore)

Input (w)	**Present State** Q_1	Q_0	**Next State** Q_1^+	Q_0^+	**FF Inputs** D_1	D_0	**Output** (y)
0	0	0	0	0	0	0	0
0	0	1	1	0	1	0	0
0	1	0	0	0	0	0	0
0	1	1	1	0	1	0	1
1	0	0	0	1	0	1	0
1	0	1	0	1	0	1	0
1	1	0	1	1	1	1	0
1	1	1	0	1	0	1	1

Table 4: State Table for Fig. 2 (Moore)

w	**P.S**	**N.S**	y
0	A	A	0
0	B	C	0
0	C	A	0
0	D	C	1
1	A	B	0
1	B	B	0
1	C	D	0
1	D	B	1

C. Equation Validation using K-Maps

K-Map for $D_1(w, Q_1, Q_0)$ (Horizontal Layout):

w \ Q_1Q_0	00	01	11	10
0	0	1	1	0
1	0	0	0	1

- **Validated Equation:** $D_1 = \overline{w}Q_0 + wQ_1\overline{Q_0}$ ✓

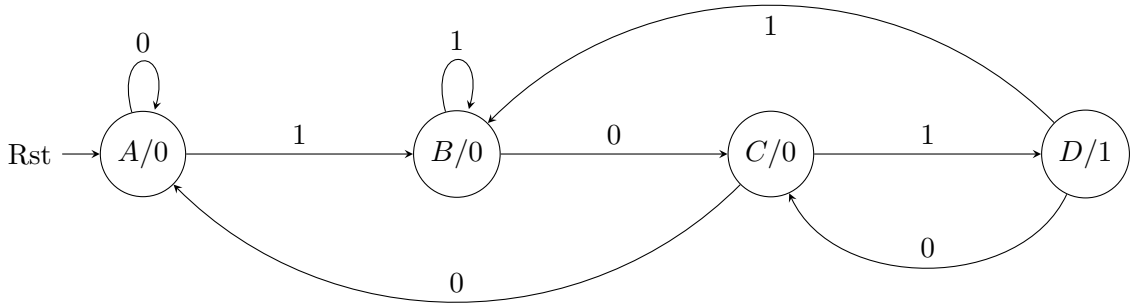
K-Map for Output $y(Q_1, Q_0)$ (Horizontal Layout):

Q_1 \ Q_0	0	1
0	0	0
1	0	1

- **Validated Equation:** $y = Q_1 Q_0$ ✓

The FSM is a Moore model which detects the sequence "101".

D. State Diagram (Moore Model Layout)



Part 2: FSM Design

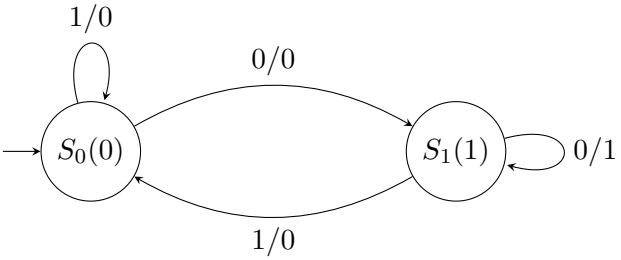
Design A: Two or More Successive Zeros Detector (Mealy Model)

1. Description & State Assignments

Detects sequence of two or more consecutive '0's.

- $S_0(0)$: Reset state / last input was '1'.
- $S_1(1)$: Single '0' detected.

2. State Diagram (Mealy)



3. Excitation Table

Format: **Input** — **Present State** — **Next State** — **FF Inputs** — **Output** (Ordered with all $w = 0$ rows first, then all $w = 1$ rows).

Table 5: Excitation Table for Design A

Input (w)	**Present State** (Q_0)	**Next State** (Q_0^+)	**FF Input** (D_0)	**Output** (y)
0	0	1	1	0
0	1	1	1	1
1	0	0	0	0
1	1	0	0	0

4. K-Map Derivations

K-Map for $D_0(w, Q_0)$ (Horizontal Layout):

w \ Q_0	0	1
0	1	1
1	0	0

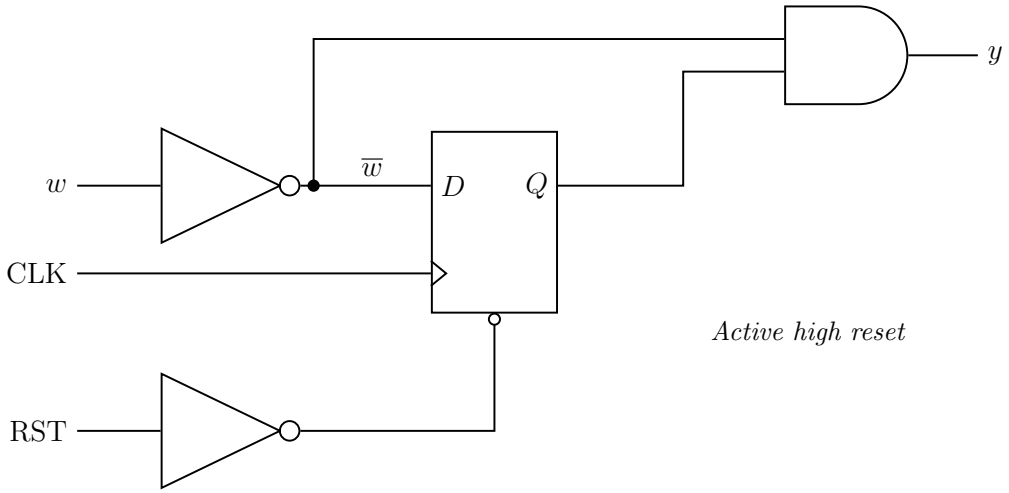
- **Result: $\mathbf{D_0 = \overline{w}}$**

K-Map for Output $y(w, Q_0)$ (Horizontal Layout):

w \ Q_0	0	1
0	0	1
1	0	0

- **Result: $\mathbf{y = \overline{w} \cdot Q_0}$**

5. Schematic Diagram

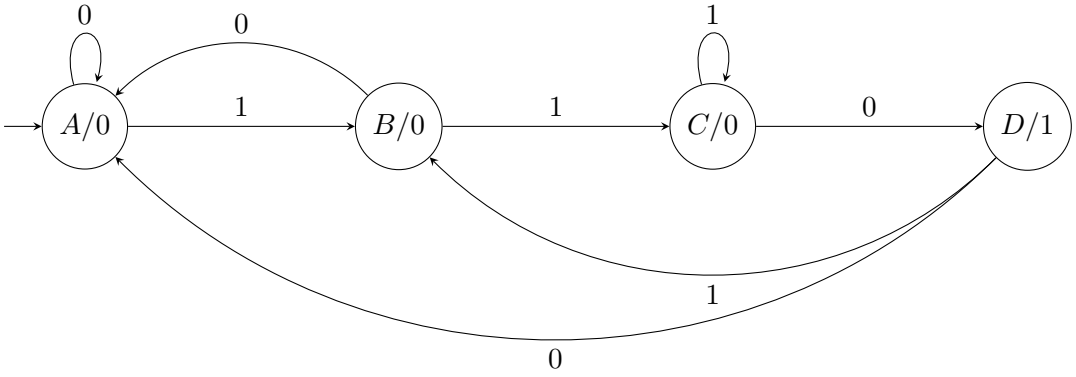


Design B: "110" Sequence Detector (Moore Model)

1. Description & State Assignments

- A: Reset state (00) / Output 0
- B: Detected '1' (01) / Output 0
- C: Detected '11' (10) / Output 0
- D: Detected '110' (11) / Output 1

2. State Diagram (Moore Layout matching Design B Notes)



3. Excitation Table

Format: **Input** — **Present State** — **Next State** — **FF Inputs** — **Output** (Ordered with all $w = 0$ rows first, then all $w = 1$ rows).

Table 6: Excitation Table for Design B

Input (w)	**Present State**		**Next State**		**FF Inputs**		**Output** (y)
	Q_1	Q_0	Q_1^+	Q_0^+	D_1	D_0	
0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0
0	1	0	1	1	1	1	0
0	1	1	0	0	0	0	1
1	0	0	0	1	0	1	0
1	0	1	1	0	1	0	0
1	1	0	1	0	1	0	0
1	1	1	0	1	0	1	1

Table 7: State Table for Design B

w	**P.S**	**N.S**	y
0	A	A	0
0	B	A	0
0	C	D	1
0	D	A	0
1	A	B	0
1	B	C	0
1	C	C	0
1	D	B	0

4. K-Map Derivations

K-Map for $D_1(w, Q_1, Q_0)$ (Horizontal Layout):

w \ Q_1Q_0	00	01	11	10
0	0	0	0	1
1	0	1	0	1

- **Result: $\mathbf{D_1 = Q_1\overline{Q_0} + \overline{Q_1}Q_0w}$**

K-Map for $D_0(w, Q_1, Q_0)$ (Horizontal Layout):

w \ Q_1Q_0	00	01	11	10
0	0	0	0	1
1	1	0	1	0

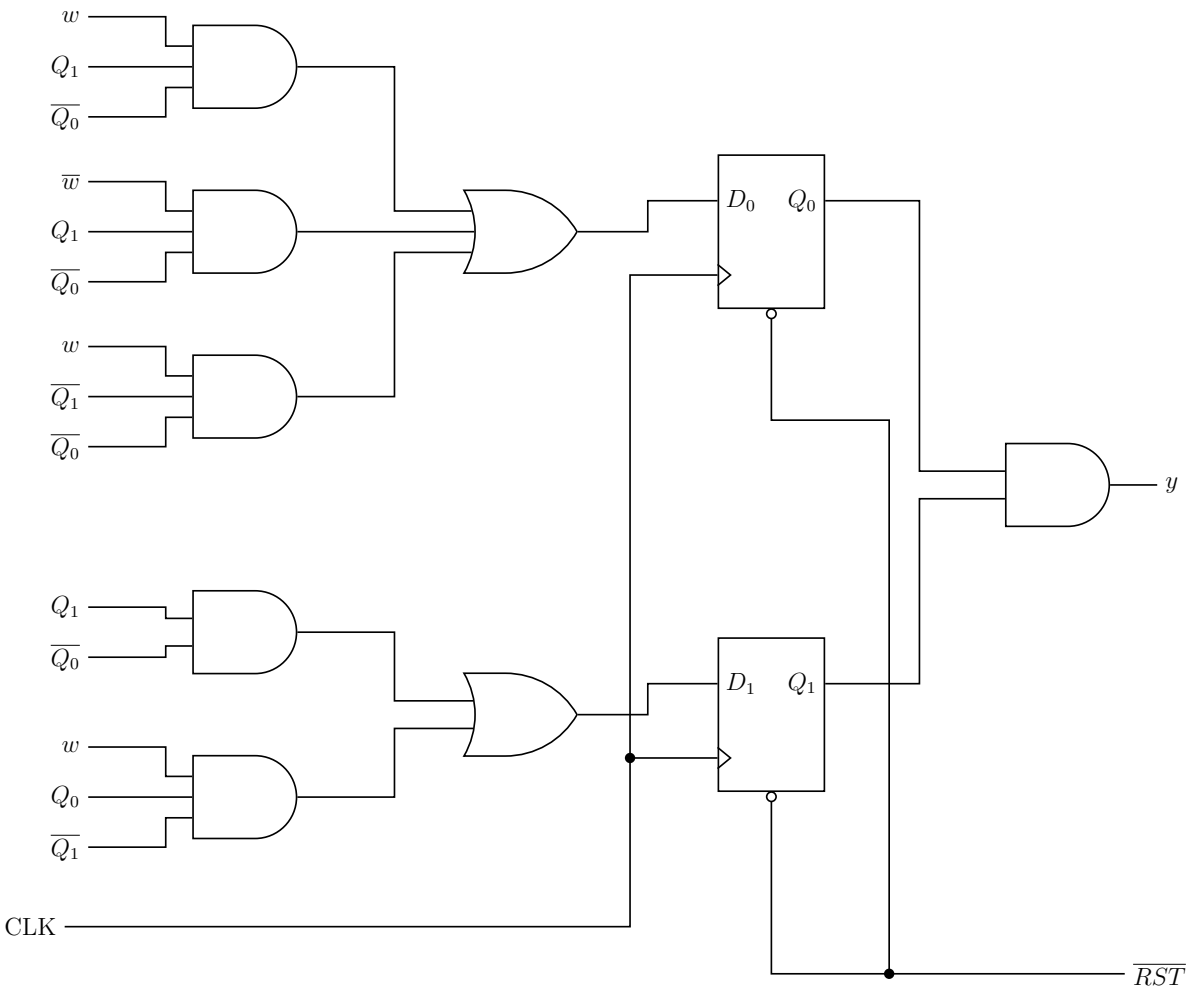
- **Result: $\mathbf{D_0 = w\overline{Q_1 Q_0} + \overline{w} Q_1 \overline{Q_0} + w Q_1 \overline{Q_0}}$**

K-Map for Output $y(Q_1, Q_0)$:

$$\mathbf{y = Q_1 Q_0}$$

5. Schematic Diagram

The circuit represents a Moore FSM using DFFs which detects the sequence "110".



Design C: "110" or "101" Sequence Detector (Moore Model - Overlapping Allowed)

1. Description & State Assignments

- A: Reset state / Empty (000) / Output 0
- B: Detected '1' (001) / Output 0
- C: Detected '11' (010) / Output 0
- D: Detected '10' (011) / Output 0
- E: Detected '110' (100) / Output 1
- F: Detected '101' (101) / Output 1
- Unused states (110 and 111) are treated as Don't Cares (X).

2. Full Excitation Table (Sorted from 0000 to 1111)

Format: **Input** — **Present State** — **Next State** — **FF Inputs** — **Output** (Listing all combinations from 0000 to 1111 sequentially based on binary value of $wQ_2Q_1Q_0$, with X for unused states).

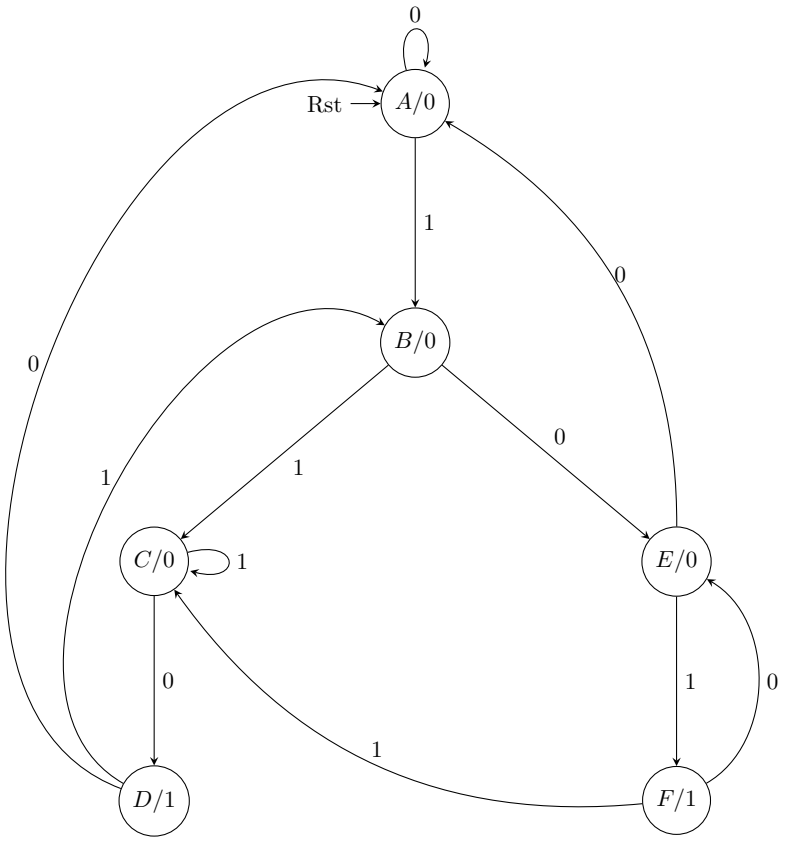
Table 8: Full Excitation Table for Design C (Sorted 0000 to 1111)

Input (w)	**Present State**			**Next State**			**FF Inputs**			**Output** (y)
	Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	D_2	D_1	D_0	
0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	1	0	0	1	0	0
0	0	1	0	0	0	0	0	0	0	0
0	0	1	1	1	0	1	1	0	1	0
0	1	0	0	0	1	0	0	1	0	1
0	1	0	1	0	0	0	0	0	0	1
0	1	1	0	X	X	X	X	X	X	X
0	1	1	1	X	X	X	X	X	X	X
1	0	0	0	0	0	1	0	0	1	0
1	0	0	1	0	1	1	0	1	1	0
1	0	1	0	1	0	0	1	0	0	0
1	0	1	1	0	1	1	0	1	1	0
1	1	0	0	0	1	1	0	1	1	1
1	1	0	1	1	0	1	1	0	1	1
1	1	1	0	X	X	X	X	X	X	X
1	1	1	1	X	X	X	X	X	X	X

Table 9: State Table for Design C ($A = 000$, $B = 001$, $C = 010$, $D = 011$, $E = 100$, $F = 101$)

w	**P.S**	**N.S**	y
0	A	A	0
0	B	C	0
0	C	A	0
0	D	F	0
0	E	C	1
0	F	A	1
X	X	X	X
X	X	X	X
1	A	B	0
1	B	D	0
1	C	E	0
1	D	D	0
1	E	D	1
1	F	E	1
X	X	X	X
X	X	X	X

3. State Diagram (Moore Layout matching Design C Notes)

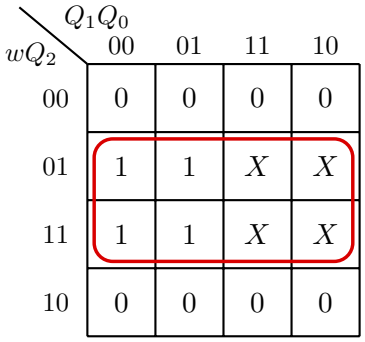


4. K-Map Derivations

Note: Unused states ($Q_2Q_1Q_0 = 110$ and 111) are marked as Don't Cares (X) in the Karnaugh map cells where $Q_2 = 1, Q_1 = 1$. Both row variables (wQ_2) and column variables (Q_1Q_0) follow

Gray code ordering $(00, 01, 11, 10)$.

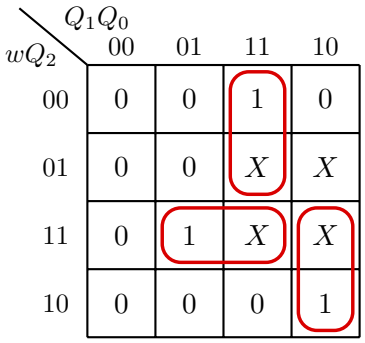
K-Map for Output $y(w, Q_2, Q_1, Q_0)$**:**



wQ_2 \ Q_1Q_0	00	01	11	10
00	0	0	0	0
01	1	1	X	X
11	1	1	X	X
10	0	0	0	0

- **Equation for** y**:** $\mathbf{y = Q_2}$

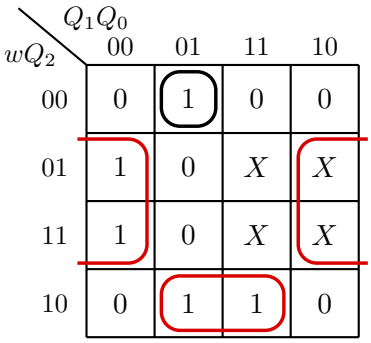
K-Map for $D_2(w, Q_2, Q_1, Q_0)$**:**



wQ_2 \ Q_1Q_0	00	01	11	10
00	0	0	1	0
01	0	0	X	X
11	0	1	X	X
10	0	0	0	1

- **Equation for** D_2**:** $\mathbf{D_2 = \overline{w}Q_1Q_0 + wQ_2Q_0 + wQ_1\overline{Q_0}}$

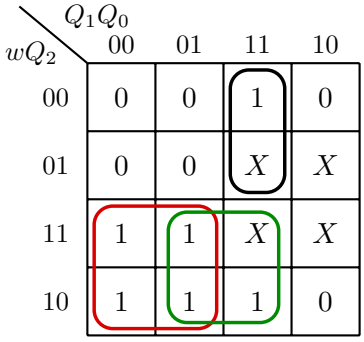
K-Map for $D_1(w, Q_2, Q_1, Q_0)$**:**



wQ_2 \ Q_1Q_0	00	01	11	10
00	0	1	0	0
01	1	0	X	X
11	1	0	X	X
10	0	1	1	0

- **Equation for** D_1**:** $\mathbf{D_1 = Q_2\overline{Q_0} + w\overline{Q_2}Q_0 + \overline{w}\overline{Q_2Q_1}Q_0}$

K-Map for $D_0(w, Q_2, Q_1, Q_0)$:



wQ_2 \ Q_1Q_0	00	01	11	10
00	0	0	1	0
01	0	0	X	X
11	1	1	X	X
10	1	1	1	0

- **Equation for D_0: $\mathbf{D_0 = w\overline{Q_1} + wQ_0 + \overline{w}Q_1Q_0}$**

5. Schematic Diagram

